

RENEWABLE CHARGING STATION USING SOLAR AND WIND TURBINE

A PROJECT REPORT for Design Thinking and Methodology

Submitted by

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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RAMAPURAM

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BONAFIDE CERTIFICATE

Certified that this project report “RENEWABLE CHARGING STATION USING SOLAR AND WIND TURBINE” is the Bonafide work of “**KEERTHAN.A.S (RA2311002020003),PURUSOTHAM.M(RA2311002020041),KISHORE.S. M. (RA2311002020047), D.CHAITANYA PAVAN KUMAR REDDY (RA2311002020057), HARI NIVAS.M.S (RA2311002020067)**” of **III Semester B.Tech. MECHANICAL ENGINEERING** during the academic year 2024-25 who carried out the project work under my supervision in **21DCS201P – Design Thinking and Methodology**.

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INTERNAL EXAMINER – I

INTERNAL EXAMINER – II

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ABSTRACT

This project presents an innovative approach to renewable energy utilization through a solar-powered street lighting and portable charging system, controlled via RFID and monitored using an IoT platform. The system harnesses solar energy to charge a dedicated battery, which powers an LED street light, ensuring sustainable, off-grid illumination. A portable charger is included in the setup, activated through RFID verification, which limits access and ensures energy-efficient usage. Once the RFID tag is scanned, the portable charger is turned on for a pre-defined duration, providing secure and controlled charging to users while preventing misuse. Integrated with the Blynk IoT platform, this system allows for real-time monitoring and management, including viewing battery status, light activation times, and portable charger usage history. The Blynk app offers remote access to key data, alert notifications, and performance analytics, enhancing user experience and operational efficiency. By utilizing solar energy, the system reduces reliance on conventional power sources, contributing to environmental sustainability and lowering operational costs. This IoT-enhanced solution addresses the growing demand for smart, energy-efficient urban infrastructure and can be implemented in various public spaces, including parks, bus stations, and rural areas where reliable power sources may be limited. With its user-friendly interface and integration of IoT for remote monitoring, this solar-powered street light and charging solution is not only energy-conscious but also aligned with modern smart city initiatives.

EXPLORE PHASE

1.1 STEEPANALYSIS

SOCIAL & DEMOGRAPHICS

- Sustainability
- Friendly ecosystem
- Encouraging renewable resources
- Impressed technology

TECHNOLOGY

- Solar panels
- Smart Features
- Battery storage
- Wind turbine

ECONOMY

- Less electricity bill
- Cost of implementing is less
- Simple working principles

ENVIRONMENT & NATURE

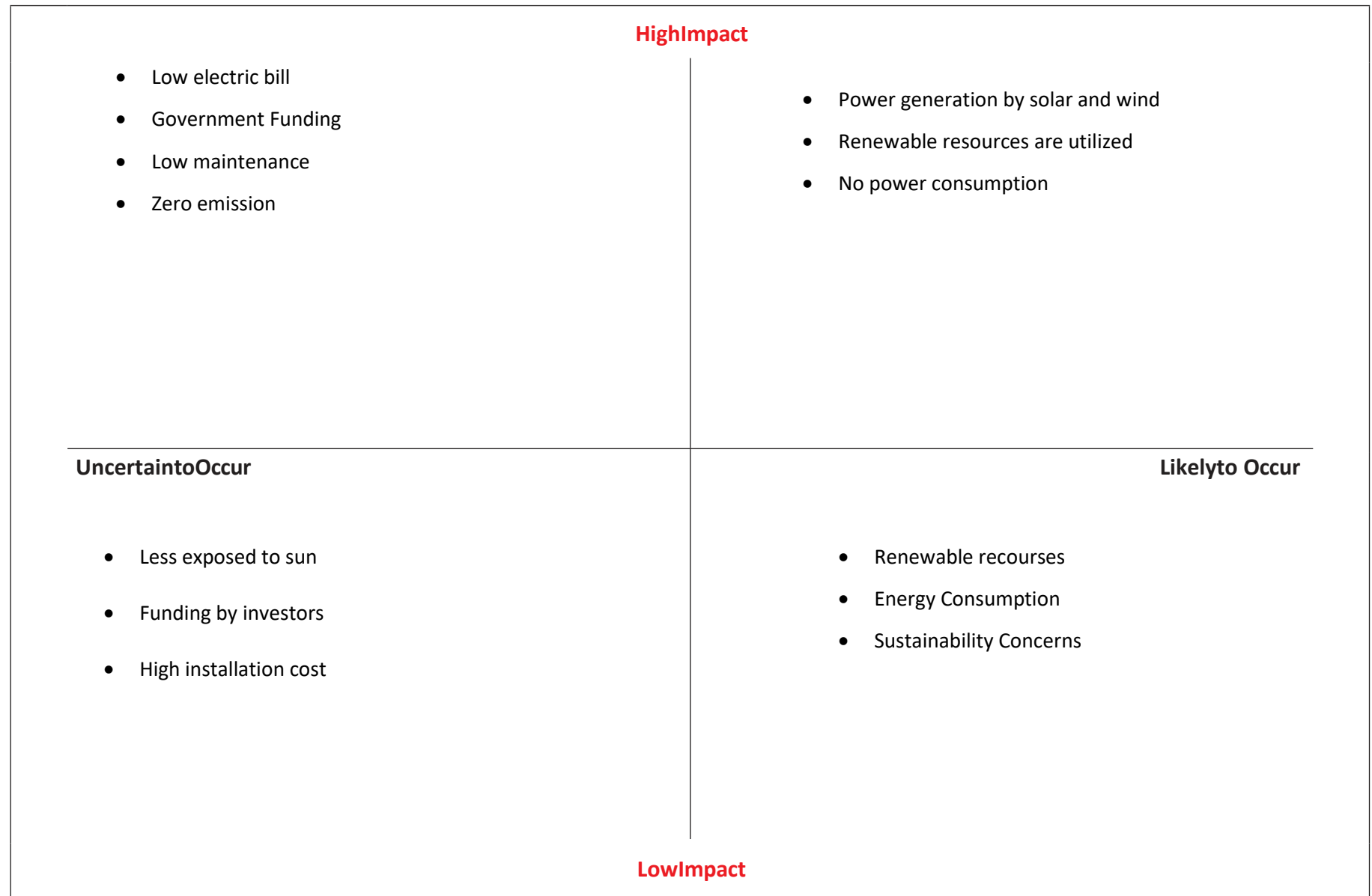
- Sustainable development
- No emission
- Renewable resources
- Less radiation (Birds)

POLITICS & LEGAL

- Fast patent creation
- Incubation cost is less
- Easier approval

**YOUR DESIGN
CHALLENGE
ISSUE**

1.2 STEEP ANALYSIS MATRIX



1.3 STEEP ANALYSIS PRIORITIZATION

Discuss 2 key trends from the High Impact–Likely to occur quadrant:

1. Low energy consumption. Therefore no electricity bills
2. There is carbon foot prints as we use only renewable resources for energy generation.

Discuss 2 trends from the High Impact–Uncertain to occur quadrant

1. Low exposed to sun need more space for efficient working.
2. Cost of solar panels is comparatively high.

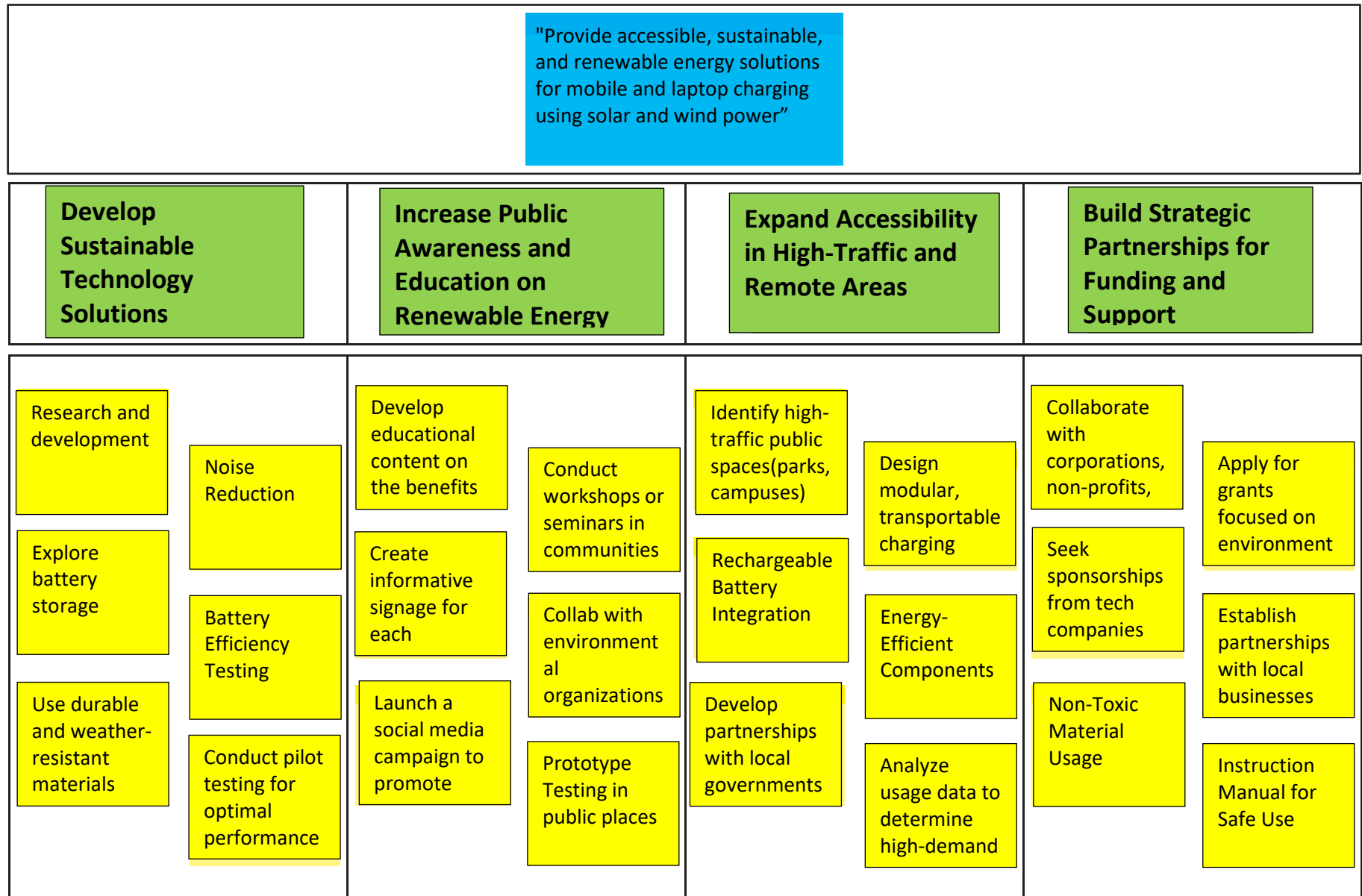
1.4 STEEP ANALYSIS MATRIX

	URGENT	LESS URGENT
IMPORTANT	Technological - Ensuring the usage of solar and wind turbine effectively to make this charging setup more efficiently. Economic – electricity bill can be reduced and low electricity consumption. Environmental – addressing high carbon emission due to thermal power plant to produce electricity.	Social – to substitute the current charging method. Political - Compliance with local and international product safety regulations for electricity generation. Environmental - Long-term strategy for sustainable development in power generation to reduce carbon footprint.
LESS IMPORTANT	Economic - Short-term cost adjustments for quick market entry. Technological - Quick design modifications based on initial user feedback to improve usability.	Social - Long-term community engagement for developing continuous charging. Political - Aligning with upcoming electrical policies that may encourage renewable technologies.

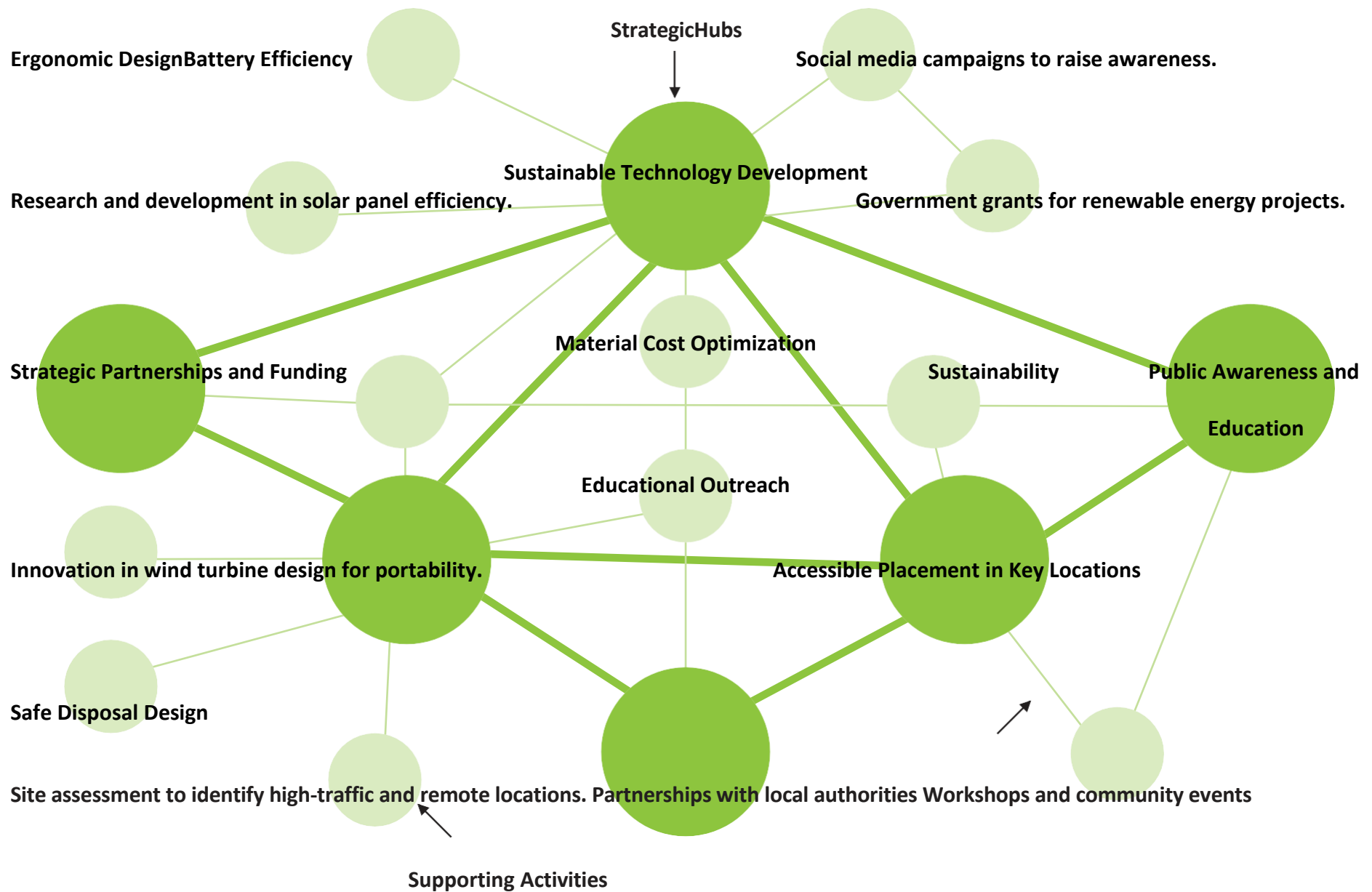
1.5 SYNTHESIS: MAKING SENSE OF STEEP ANALYSIS AND STRATEGIC PRIORITIES

Assessment Questions	Synthesis : Sense Making
What relationships among the trends do you perceive? How are they related? Why are these relationships important	There is a strong relationship between environmental sustainability, technological advancements, and consumer demand for eco-friendly solutions. As awareness of climate change grows, there's an increasing trend toward adopting renewable energy sources, especially in areas of mobile and portable charging.. These relationships are crucial because they show a growing market for green energy solutions and reinforce the importance of reducing carbon footprints.
What opportunities and/or challenges need immediate attention going forward for your design challenge? And why?	Opportunities include the potential to tap into the expanding market of environmentally conscious consumers, especially in public spaces, parks, and remote areas where access to electricity is limited. Challenges involve ensuring consistent power output despite variability in sunlight and wind conditions, as well as managing initial installation and maintenance costs. Addressing these challenges is vital to create a reliable, user-friendly solution that can attract adoption and trust from users.
What would it take to create positive change on this issue relating to your design challenge?	To create positive change, the project would need to focus on cost-effective and robust technology that can withstand diverse environmental conditions. Collaboration with local governments and organizations to place these stations in strategic locations would increase accessibility and visibility, raising public awareness about renewable energy. Additionally, user education on the benefits of using renewable charging stations over traditional methods could further drive positive behavioral shifts towards sustainability.
Who else would be interested in this issue? Why should they care? What conversations would you have with them?	Stakeholders include environmental organizations, tech companies, government agencies, and educational institutions. Environmental groups would be interested in promoting sustainable practices, while tech companies may see potential for product partnerships or sponsorships. Governments would value the environmental benefits, especially in urban development projects, and educational institutions could use this as an example of practical sustainability for students. Conversations would focus on collaboration opportunities, shared goals in sustainability, and ways to expand the project's impact and reach

1.6 MAPPING ORGANIZATION ACTIVITY

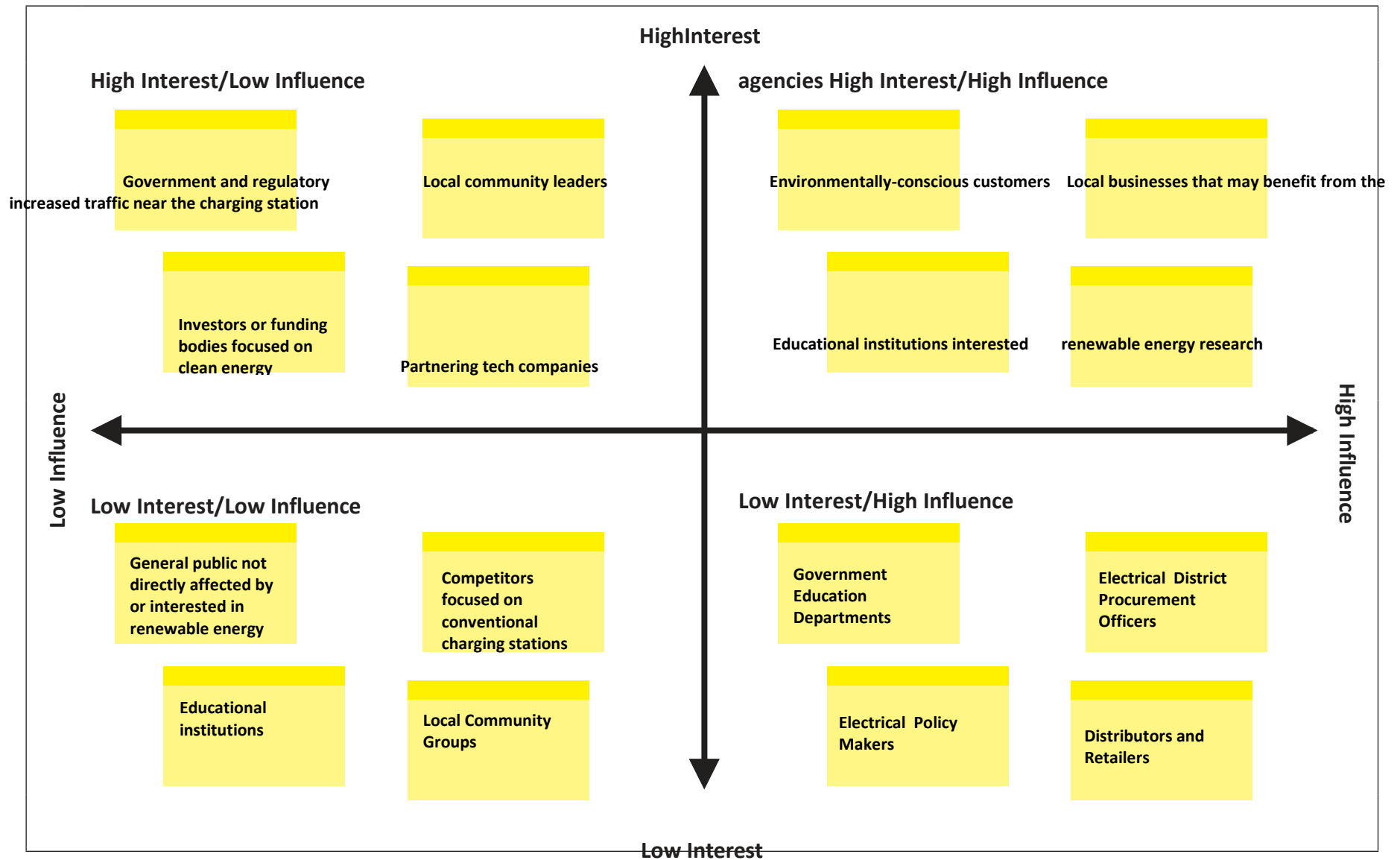


1.7 KEY COMPONENTS OF ACITIVITY



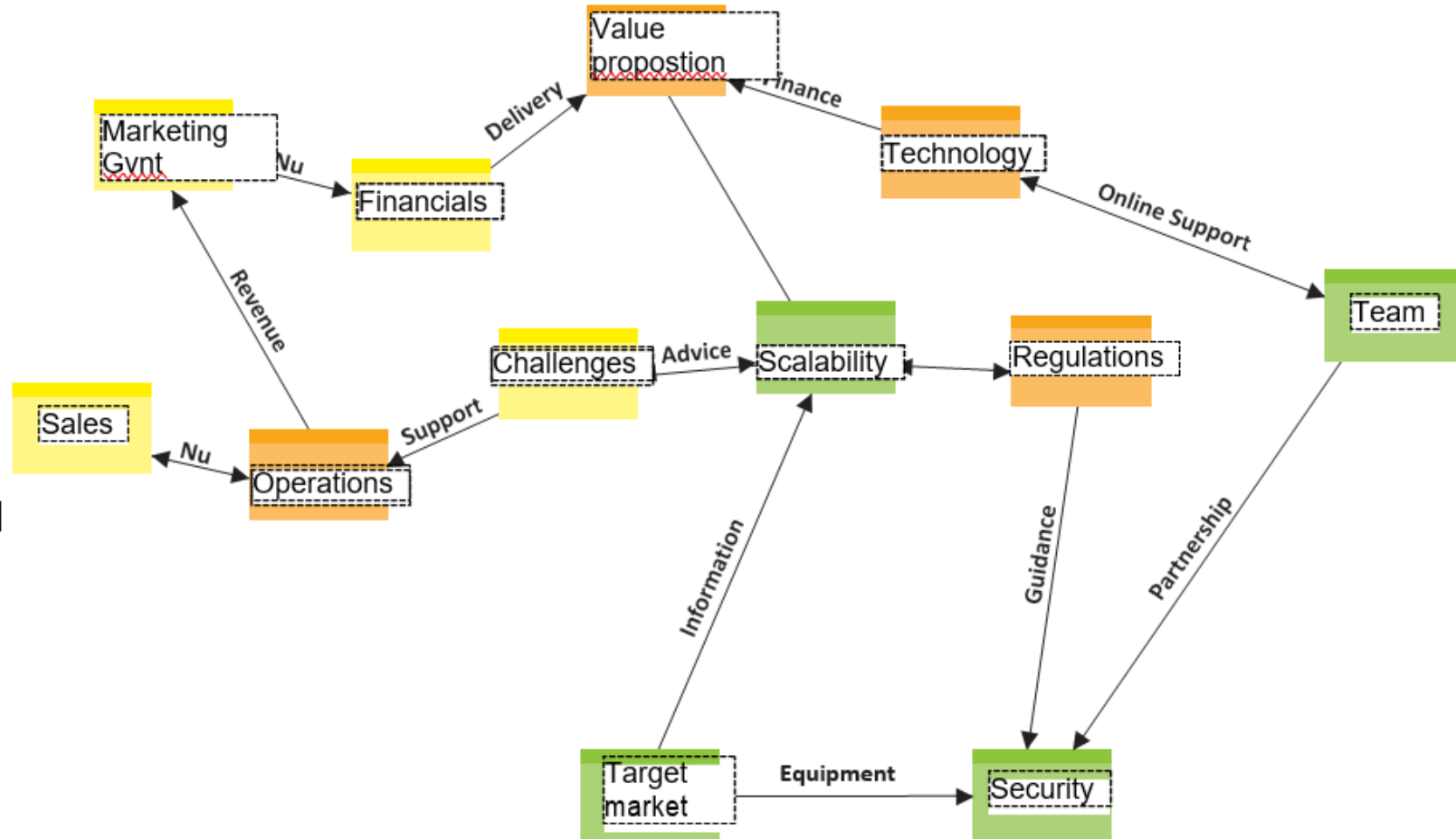
1.8 STAKEHOLDER MAPPING MATRIX

(Identify your relevant key stakeholders)



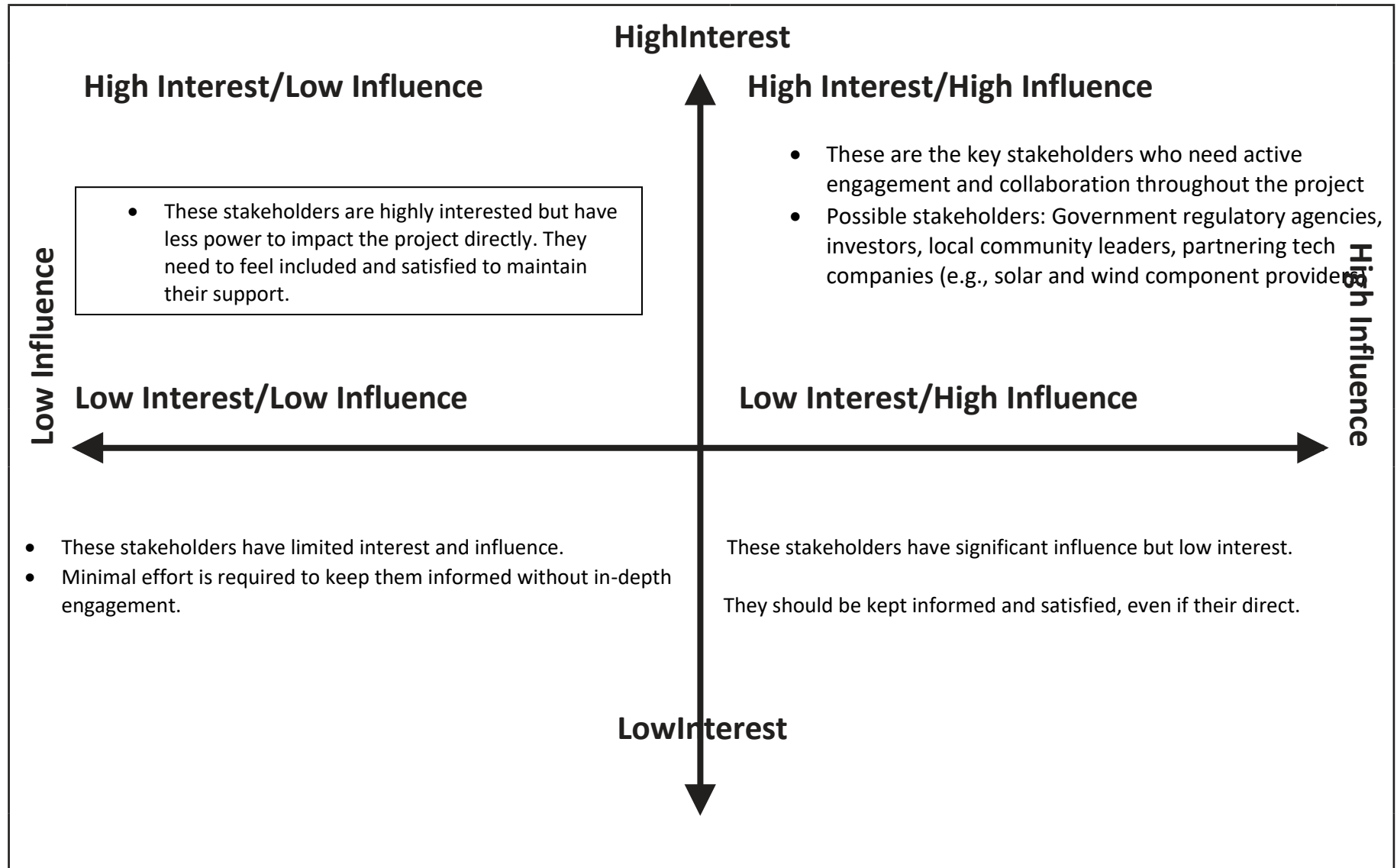
1.9 STAKEHOLDER LINKS & RELATIONSHIP MAPPING

(Definethelinksandrelationshipsbetweenstakeholders)



1.10 STAKE HOLDER PRIORITY MAPPING MATRIX

(Develop engagement Strategies)



1.11 STAKE HOLDER ANALYSIS & ENGAGEMENT

Key Stakeholders	Relationships	Stakeholder's Interest(s) in the Design Challenge	Impact Assessment	Strategies to Gain Support or Reduce Obstacles
Government/Regulatory Agencies	Regulatory and policy oversight	Ensuring compliance with renewable energy standards and local regulations	High - approval needed for project progress	Regular updates on project, ensure compliance, address concerns
Investors/Funding Bodies	Financial backing	Financial return on investment, innovation in renewable tech	High - fund availability and expectations	Transparent reporting on project milestones, ROI updates
Local Community Leaders	Local influence and public opinion	Job creation, environmental impact, community benefits	Medium to high - community acceptance critical	Community meetings, engagement workshops, address environmental impact
Partnering Tech Companies	Collaboration for tech components	Showcasing new tech, successful implementation of components	High - component supply and tech success	Align goals, establish joint development plans, regular coordination
Environmentally-Conscious Consumers	Potential users	Access to sustainable charging solutions	Medium - potential customer base	Social media campaigns, eco-friendly branding
Utility Companies	Grid access and energy supply	Ensuring grid stability, possible revenue from grid access	Medium to high - essential for integration	Meetings to align on energy needs, address grid impact
Educational Institutions	Collaboration for research, awareness	Research opportunities, project case study	Low to medium - support but limited influence	Offer site tours, research collaborations, data sharing
General Public	End users and general community sentiment	Convenience, minimal disruption to daily life	Low - indirect beneficiaries	Basic public notices, minimal impact assurance
Competitors	Market players with similar products	Tracking competition and differentiation	Low - influence only on market competitiveness	Market differentiation strategy, avoid direct engagement

1.12 PROJECT BRIEF AND OPPORTUNITY FRAMING

A.CHALLENGEORPROBLEMDEFINITION		
ProjectSponsor	OrganizationName	SRM IST
	AddressandContact	Ramapuram, Chennai, Tamil Nadu
	ContactPerson(s)	Mani (87XXXXXXX9)
ProjectTitle	Renewable charging station using solar and wind turbine for mobile and laptop charging	
DesignChallenge	Develop a reliable, portable, and eco-friendly charging solution for mobile devices and laptops, powered by renewable energy sources such as solar panels and wind turbines.	
DesignChallengeContextandBackground Info	The rising use of mobile devices and laptops demands a convenient and sustainable charging solution, especially in remote areas or during travel where traditional power sources are unavailable. The challenge is to harness renewable energy sources, specifically solar and wind, to meet this demand while reducing reliance on fossil fuels. This approach aligns with global sustainability goals and provides an alternative for users looking to minimize their carbon footprint.	
	This design challenge aligns with the organization’s mission to advance sustainable technology. By creating a renewable-powered charging solution, the organization can strengthen its position in the green technology market, cater to environmentally-conscious consumers, and contribute to global environmental conservation efforts.	

1.12 PROJECT BRIEF AND OPPORTUNITY FRAMING

A. CHALLENGE OR PROBLEM DEFINITION(contd...)	
Goals and objectives of the Design Challenge	Design thinking methodology using solar panels and wind turbine to charge mobiles and laptop Reduce dependency on conventional electricity sources, promoting sustainability and reducing carbon emissions. Enhance accessibility to green energy solutions, enabling users to charge devices anytime and anywhere.
Target users of this Design Challenge	Travelers, campers, and outdoor enthusiasts who need charging solutions in remote areas. Remote workers, students, and digital nomads seeking sustainable alternatives for device charging. Environmentally conscious consumers looking for eco-friendly technology options..
Other key stakeholders	Environmental organizations promoting green energy. Technology and hardware developers involved in portable energy solutions. Local governments and sustainability advocates interested in renewable energy solutions..
Previous efforts to solve this Design Challenge	Previous attempts have included portable solar chargers, but they are often limited by weather conditions. Some hybrid chargers exist but lack efficiency in low wind or low sunlight conditions. Limited integration of compact, dual-energy (solar and wind) solutions that are efficient and affordable for consumers..
Current activity system and process	Typical portable chargers rely solely on one renewable source, either solar or wind, limiting their effectiveness in varying conditions. Consumers often depend on power banks that require frequent charging from traditional power sources, limiting their sustainability. Lack of a fully integrated system that combines both solar and wind for consistent power generation in portable devices..

1.13 REFRAMINGTHEOPPORTUNITIESTEMPLATE

B.OPPORTUNITYFRAMING	
RealissuesbehindthisDesign Challenge	Limited access to power in remote or rural areas, impacting connectivity and productivity. Environmental impact of relying on traditional power sources, which contributes to carbon emissions. Unpredictable weather conditions affecting the reliability of single-source renewable chargers (e.g., solely solar or wind).
Inspirationsfromothersinsolving this Design Challenge	Solar-powered phone chargers used by hikers and travelers. Wind turbines designed for off-grid camping, though usually not portable or combined with solar solutions. Multi-source renewable energy systems used in larger-scale applications, adapted for portable, personal use.
Teamscontributions	Design a compact, dual-powered charging device that seamlessly integrates both solar and wind power. Innovate an adaptable power generation system that adjusts based on environmental conditions (e.g., prioritizing solar on sunny days and wind during breezy conditions). Optimize the device for portability, durability, and ease of use for a wide range of users.
Successcriteria	Ability to charge a standard mobile device within 1-2 hours using renewable energy. Device functionality across different environmental conditions (day/night, sunny/cloudy, windy/calm). Positive user feedback on usability, convenience, and eco-friendliness. Measurable reduction in the user's dependency on traditional electricity sources for device charging..
"HOWMIGHTWE"Opportunity/ possibility statement	How might we create a reliable, portable device that harnesses both solar and wind energy to charge mobile devices and laptops, offering a sustainable solution for users in various environments?

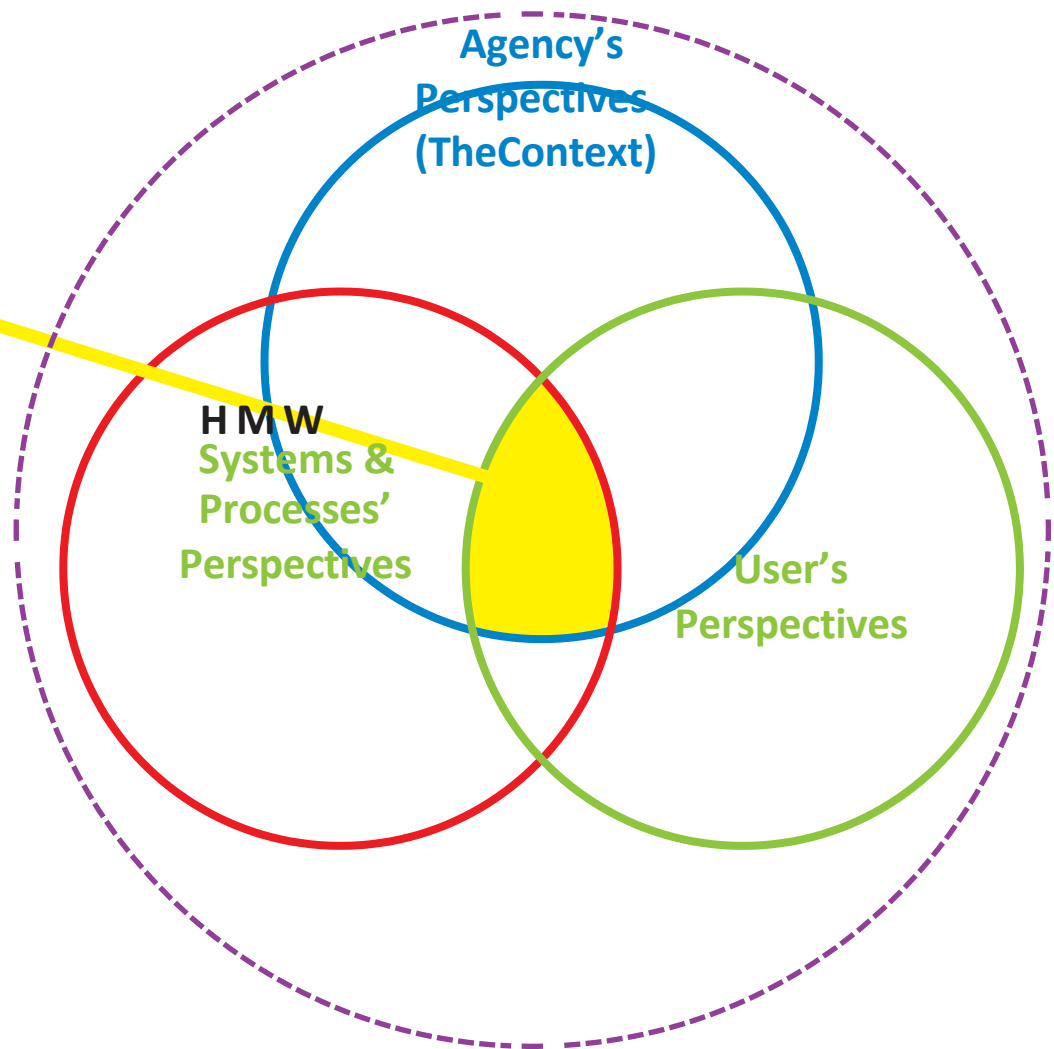
1.13 REFRAMINGTHEOPPORTUNITIESTEMPLATE

Reframeyouroportunities
and filter one

“Howmightwe ?”

- How might we ensure the device can operate reliably in diverse environmental conditions (e.g., varying sunlight and wind levels)?
- How might we design the solution to align with environmental regulations and promote sustainable energy usage?
- How might we make the charging solution accessible and affordable for broader consumer adoption?

STEEP Trends &Analysis

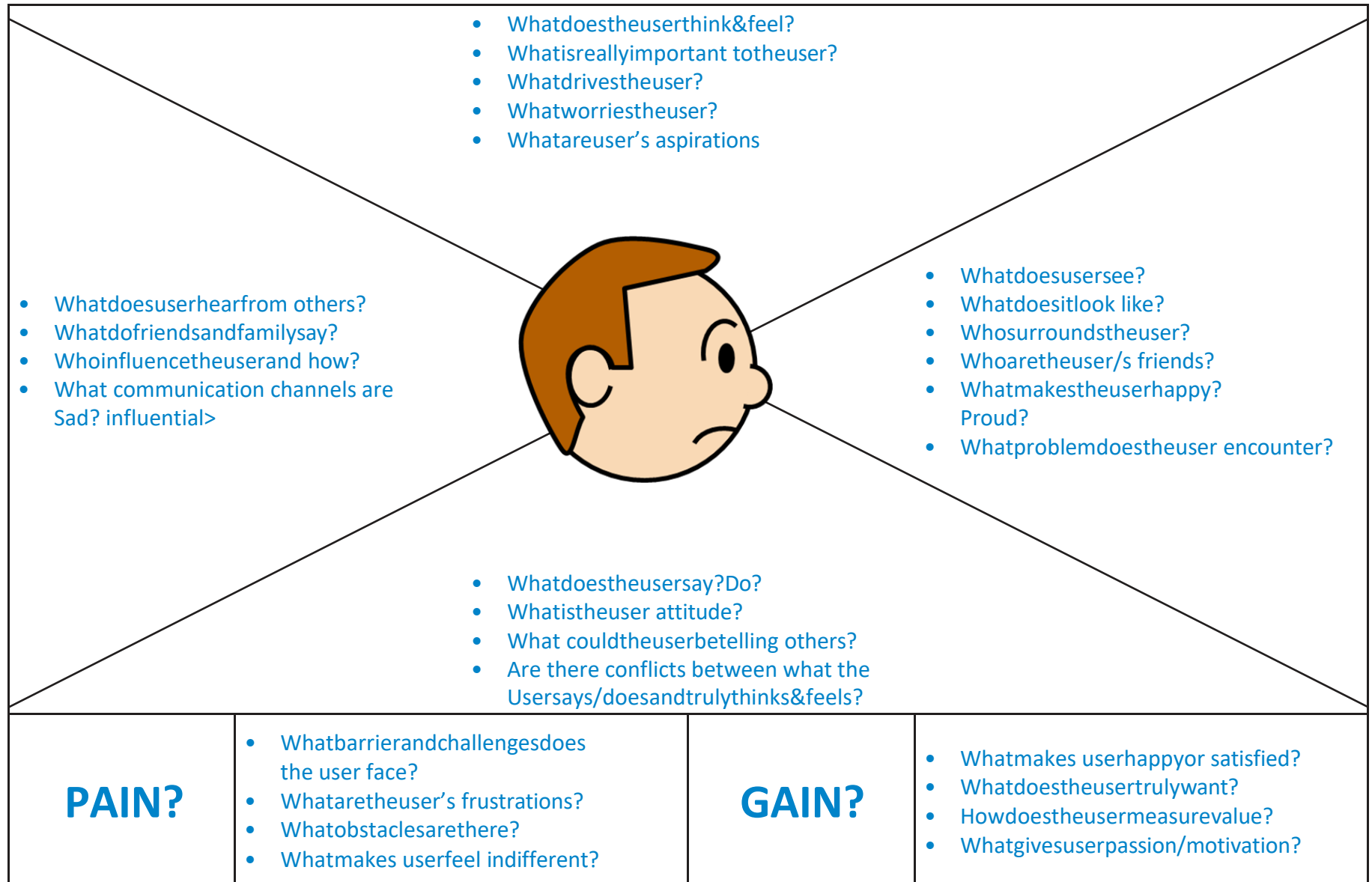


**EMPATHISE
PHASE**

2.1 POEMS FRAME WORK TEMPLATE

Location:		Date/Time/Period:
Report By:		
POEMS Framework	Field Visit and Onsite Observation	Insights
People: Who are involved? (E.g. commuters, street vendors, office workers, children, motorists, delivery men, etc.) What roles do they play? How are the people engaging with each other? How are they related? What are the relationships? What is the social context?	Identify who will use the charging station (e.g., students, office workers, travelers).	Consider how they interact with each other while using the charging station. Are there queues, social interactions, or shared usage patterns? Observe their charging needs and how long they tend to stay at the station
Objects: What artifacts are important? (E.g. bus stops signage, traffic lights, benches, etc.) What roles do they play? How are people engaging with the objects, and with their surrounding? What is interaction? How does the object influence people's behavior?	Focus on the charging station, solar panels, wind turbines, and related equipment.	Think about how these objects are used by the people. Do users need instructions, or is the design intuitive? Consider the station's visibility, accessibility, and interaction with nearby structures or objects (e.g., benches, canopies).
Environment: Where is the action taking place? (e.g. public bus, road side, government office, garden, etc.) What is happening? What are the people doing? How do the people behave in this environment? How does the environment influence people's behavior? What is the mood? Ambience?	Observe the location where the station is placed (e.g., parks, bus stops, campuses).	Document environmental factors like sunlight availability, wind patterns, and weather conditions. These are essential for optimizing renewable energy. Note the ambience and how people behave in this space □ are they relaxed, in a rush, or engaging with others?
Messages & Media: What are the messages and communication media used? (e.g. signage, online materials, posters, apps, etc.) What roles do they play?	Look at signage, instructions, and information provided at the station (e.g., how to use, charging times, environmental benefits)..	Consider if digital media, like an app or online platform, could enhance the user experience (e.g., showing available charging spots or current power levels).
Services: What are the services and support systems provided? (e.g. registration, library services, hands-on guide, online booking, etc.)	Think about the support systems available, such as technical support, maintenance services, or apps for monitoring charging.	Evaluate how services like user guides, online booking of charging spots, or live energy usage data might improve user satisfaction.

2.2 GENERATEINTERVIEWQUESTIONS



2.3 EMPATHY MAP & USER JOURNEY TO GENERATE INTERVIEW

DOING	SEEING	HEARING/SAYING	FEELING/THINKING	FRUSTATION	NEEDS/WANTS
• Document when and how users charge their devices (e.g., during breaks, while waiting for transportation). • Understand how they interact with current charging solutions, and what prompts them to use a charging station..	• Note what users observe around the charging station (e.g., other people using it, any signage). • Identify visual cues that might impact their decision to use or trust the charging station.	• Capture any feedback they share verbally or overhear from others regarding the charging station. • This could include comments on the stations reliability, convenience, or environmental impact.	• Explore users' emotions and thoughts about using renewable energy sources for charging. • Consider potential concerns, such as safety, charging speed, or scepticism about the station's power source.	• Identify pain points, like insufficient charging spots, unclear instructions, or poor power availability. • Use these insights to address common issues in your design.	• Highlight users' essential needs (e.g., fast charging, reliable power) and any additional features they desire (e.g., USB and power outlets, weather protection). • This can help prioritize features that enhance the user experience

USER INTERVIEW NOTES

InterviewerName	Pavan
NoteTaker& Observer	kumar
IntervieweeName	Prakash
Intervieweecontactdetails	7874003701
Date/Timeof Interview	27/08/24
InterviewLocation	Chennai, Tamil Nadu
Interview'sbackgroundinformation	
Gender	Male
AgeRange	25-30
Profession	Professor
EducationBackground	M.tech Mechanical
Family	Single
Hobbies	Reading, Problem Solving
Likes/Dislikes	Likes: Music Dislikes: Social Gatherings
InterviewNotes(Verbatim)	Observations/Quotes
●Background and Frequency of Use - Record general information about the user, such as their age range, profession, and educational background. - This will help understand the user segments that will most benefit from the charging station.. - Note the types of devices they use frequently and their charging needs. - This information can inform the design, such as including multiple types of charging ports. - Capture direct quotes, behaviours, and any feedback related to renewable energy charging stations. - Use this information to understand user expectations and uncover unique insights.	●User Behaviour with Current Dusting Solutions Document users' hobbies and lifestyle preferences, especially any that suggest a preference for sustainable technology. Insights here might guide decisions about aesthetics, functionality, or placement of the station

POST INTERVIEW DISCUSSION : ABOUT THE INTERVIEWEE

Interviewee's Goals &/or Motivation - Identify what drives the interviewee's interest in renewable energy or public charging stations. Are they motivated by sustainability, convenience, or cost-saving? - For instance, some users might value environmental benefits, while others focus on reliable charging access.	Interviewee's Aspirations: - Document any long-term desires related to technology, convenience, or sustainability. - This can guide the design to meet aspirational needs, such as contributing to environmental impact or enhancing urban accessibility.
Interviewee's Current Experience - Record the user's typical experiences with public charging options or renewable energy. - Understanding current usage patterns or frustrations can pinpoint gaps your design should address, like reliability or availability issues.	Interviewee's Challenges &/or Pain Points - Note specific problems the interviewee faces, such as charging accessibility, slow speeds, or difficulty finding renewable options. - These pain points will be valuable to address directly in your charging station design.
3 most memorable things about the interviewee: Capture any unique traits or standout comments from the interviewee that can inform user personas or guide specific design decisions	User Insights &/or Needs - Summarize the core insights, such as the need for quick access, ease of use, clear instructions, or weather protection. - These insights help define the primary requirements for a user-centered renewable charging station.

What went well with the Interview ? - Reflect on successful aspects, such as the relevance of the questions, the comfort level of the interviewee, or insights gained. - For example, if the interview uncovered detailed pain points, this might be a strength to replicate.
What did not go well with the Interview ? - Note any challenges, like unclear questions, limited responses, or missed insights. - Identifying areas of improvement helps refine future interviews, making them more effective.
How can we do better for next Interview?

POST INTERVIEW DISCUSSION : ABOUT THE INTERVIEWEE

- Together, these templates support a structured approach to user research in the design thinking process. They allow you to gather insights, identify areas for improvement, and clearly communicate findings to others involved in designing the renewable charging station. By using these templates, you ensure that user needs and challenges remain central to your design efforts, increasing the chances of creating a product that resonates with and effectively serves its users.
- Highlight memorable insights, like specific needs or unique perspectives, which can guide the design process

END OF INTERVIEW NOTES

POST INTERVIEW DEBRIEF PRESENTATION:

Summary proof interviewee

	InsertPhotoHere		Overview of interviewee (Summarize basic demographic details such as age, occupation, location, and any other relevant information like lifestyle, hobbies, or habits that might influence their charging needs.

Summary profile of interviewee

Goals/Motivation: • Identify what the interviewee hopes to achieve with a portable charging solution. Common goals might include reducing reliance on non-renewable power sources, ensuring their devices stay charged while on the move, or contributing to environmental sustainability..	Aspiration: Capture the interviewee's long-term aspirations related to sustainable living or energy independence.
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Current Experience: Document the interviewee's current experience with charging devices, including any challenges they face when trying to charge their devices on the go.	Challenges and Pain Points: • Identify specific pain points, such as limited access to charging options, unreliable power sources, or environmental concerns about using fossil fuels..
3 most memorable things about the interviewee: • Summarize the most memorable or impactful things shared by the interviewee, highlighting their deep needs and insights related to renewable charging.	User Insight & Deep Needs: • Identify what the interviewee hopes to achieve with a portable charging solution. Common goals might include reducing reliance on non-renewable power sources, ensuring their devices stay charged while on the move, or contributing to environmental sustainability •

DOCUMENTING INSIGHTS & NEEDS: EXAMPLE

INSIGHT #: • Write each key insight you've gathered from the interview, focusing on the underlying needs and motivations related to renewable charging.

Example: "Users are often unable to charge their devices while hiking or camping, which creates a desire for a lightweight, durable renewable charging solution cleanliness and personal comfort."

Need:

- Clearly state what the interviewee needs, based on the insight. In this example, it might be "A portable and weather-resistant charging solution that can be used in remote outdoor locations."

Quotes1:

"I really need something lightweight that I can take anywhere, especially on multi-day hikes."

Quotes2:

"It would be amazing if I didn't have to worry about power even in the middle of nowhere."

PERSONACANVAS		Persona Name: Mr. Varma
Demographic Profile: Age: 28 Gender: Male Home: Chennai Family: EducationBackground: M.tech Mechanical	Goals: Identify what this persona hopes to accomplish with a renewable charging	Deep Need Statement: "Create a concise statement summarizing the persona's primary need, like: "Outdoor Enthusiast Emma needs a reliable, eco-friendly charging solution that works in diverse weather conditions, so she can stay connected and safe without harming the environment."
	Motivation/Aspiration: Describe what motivates this persona to seek renewable charging options. This might be a desire for sustainability, convenience, or energy independence.	
Hobbies/Likes/Dislikes: Hobbies: Reading Likes: Music Dislikes: Social Gatherings	Challenges/Pain Points: Note specific problems the persona faces, such as access to power in remote areas or the inconvenience of carrying heavy battery packs.	
Social &Family Lifestyle: The man typically works in a power plant setting, interacting daily with workers and clints. Their professional reputation is important to them, influencing how they present themselves in the powerplant.	Behavior: The man tends to be proactive in finding solutions to improve power conditions often researching new tools or methods to enhance the workers environment.	

EXPERIMENT PHASE

3.1 SCAMPER WORKSHEET

S	Substitute	<i>(What can you substitute? What or who could you use instead?)</i> - Consider substituting materials or components with eco-friendly or durable alternatives. For instance, use recycled or biodegradable materials for the station casing. - Substitute traditional battery storage with more sustainable options like supercapacitors or other advanced storage technology..
C	Combine	<i>(What can you combine or bring together?)</i> - Combine solar and wind energy generation to increase efficiency and reliability. - Integrate additional functionalities, like a USB hub or wireless charging pad, to support different types of devices.
A	Adapt	<i>(What can you adapt for use as a solution?)</i> - Adapt the design for different environments, such as urban parks, hiking trails, or beach locations. This could mean adding portability features or making the unit weather-resistant. - Use modular design to adapt the charging station for various power needs, allowing users to charge one or multiple devices.
M	Modify	<i>(Can you modify or change the item in some ways?)</i> - Modify the shape, size, or configuration of the station to make it more ergonomic and easier to transport. - Explore modifying the energy capture method, such as using foldable solar panels for compactness or retractable wind turbines for ease of transport..
P	Put to other uses	<i>(What other or new ways could this be used?)</i> - Consider other applications, like using the station as an emergency power source or a lighting station in remote areas. - Think about how this station could also serve as an educational tool, demonstrating renewable energy in action.
E	Eliminate	<i>(What can you get rid of?)</i> - Remove unnecessary components or steps in assembly to streamline the design and reduce costs. - Minimize the number of moving parts to reduce the risk of mechanical failure and increase durability.
R	Reverse	<i>(What could you rearrange or change around?)</i> - Reverse the approach by focusing on a specific power source (either solar or wind) depending on the primary environment, rather than a combined approach. - Instead of users coming to the charging station, explore a model where the station is more mobile or personalized, like a portable solar/wind charger that users can carry with them.

3.2 RECONNECTING WITH OUR PERSONA

Behaviors	Aspirations	Motivations	Challenges	PainPoints
Observe and document how users typically use their mobile devices and laptops in various locations. Note how frequently they charge their devices, where they face challenges accessing power, and if they have any routines related to charging.	Identify user aspirations, such as the desire for independence from traditional power sources, commitment to eco-friendly solutions, or a need for reliable power in remote or outdoor settings.	Understand what motivates users to seek renewable charging options. For instance, they may be motivated by sustainability concerns, the convenience of portable power, or a need for reliable charging solutions when they are outdoors or off-grid.	Document the common challenges users face, like the lack of charging options in remote areas, the inconvenience of carrying heavy power banks, or the inability to charge multiple devices simultaneously.	Identify pain points such as the need for long charging times, device overheating in direct sunlight, or the limited power output of small portable solar panels.

User Insights

- Summarize key insights from your research, focusing on user needs and behaviors. For example, "Users need a lightweight, weatherproof charging solution that can provide reliable power even on cloudy days or in low-wind conditions."

DeepUserNeeds

- Articulate the deep, underlying needs that the charging station should address. This could include "a sense of energy independence" or "a sustainable power source that doesn't contribute to environmental degradation."

Constrains&/orDesignCriteria

- Outline any specific constraints for the design, like cost limitations, weight restrictions, or durability requirements. Also, consider environmental constraints, such as minimal impact on natural habitats or minimal waste production.

**ENGAGE
PHASE**

4.1 STORY BOARD

Directions:

- Review your persona ,the ideal user experience journey and the prototype you have created.
- Brainstormhowyouintendtoillustrateyourpersona'sstoryinacoherentmannerincludingthenarratives.
- Use the story board template/canvas provided and start drawing.

Title: "Power on the Go: A Renewable Charging Journey"

Panels: • Use the panels to illustrate key moments in the user's interaction with the charging station. This might include: 1. Initial Encounter: Show the user discovering the charging station in a natural setting (like a park or hiking trail) or on their device (if there's a mobile app involved). 2. Set-Up: Depict the process of setting up the charging station, such as adjusting the solar panels or activating the wind turbine. 3. Charging Experience: Illustrate the user plugging in their mobile or laptop and feeling confident in the renewable, sustainable power source. 4. Completion: Show the user's device fully charged, with the user feeling satisfied, empowered.	Narrate: This template is designed to outline a narrative of the user experience journey, focusing on the ideal story for your renewable charging station solution.	First Use A portable charger is included in the setup, activated through RFID verification, which limits access and ensures energy-efficient usage. Once the RFID tag is scanned, the portable charger is turned on for a pre-defined duration, providing secure and controlled charging to users while preventing misuse.
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4.2 STORYBOARDING CANVAS

BEGINNING		MIDDLE END		
The Personal	The Setting	The Problems	The Solutions	The Resolutions
Describe the user persona in detail. For instance, “Emma, an outdoor enthusiast and environmental advocate, often struggles with keeping her devices charged during multi-day hikes.”.	Set the scene where the user encounters the charging station. For example, “Emma is on a remote hiking trail with limited access to traditional power sources.”.	Identify specific pain points that lead the persona to the solution. For instance: “Emma’s phone battery is nearly drained, but she needs it for navigation and safety.” “She feels frustrated because carrying multiple power banks is heavy and not eco-friendly.”	Show how the renewable charging station addresses these problems. Examples: “Emma spots a renewable charging station along the trail, powered by both solar panels and a small wind turbine.” “She quickly connects her phone to the station and appreciates the eco-friendly power source.”	Describe how the charging station positively impacts the user’s experience. For example: “Emma feels reassured and grateful for a convenient, sustainable way to charge her devices.” “She continues her hike knowing she can rely on renewable energy for power, aligning with her environmental values.”

EVOLVE PHASE

5.1 STRATEGIC REQUIREMENT TEMPLATE

StrategicRequirements	- Solar panels to capture sunlight and convert it into electrical energy. - Wind turbines to harness wind energy, supplementing the solar panels on cloudy days or at night.		
KeySolutionComponentsof the Big Idea	Renewable Energy Generation Solar panels to capture sunlight and convert it into electrical energy. Wind turbines to harness wind energy, supplementing the solar panels on cloudy days or at night.	Energy Storage System Battery storage to maintain power supply during low sunlight and low wind periods	Charging Infrastructure Charging ports and cables compatible with various devices, including mobile phones and laptops. Safety systems to regulate power output and prevent overcharging.
CapabilitiesRequiredto Deliver this Solution Component	Expertise in renewable energy systems, particularly in small-scale solar and wind power technologies.	Ability to design and develop energy storage solutions that are safe and durable	Knowledge of charging infrastructure technology and device compatibility.
Current Organizational Assets&capabilitiestobe Leveraged	Existing Infrastructure: Any current experience in renewable energy installations.	Technical Expertise: In-house engineering or tech teams that may contribute to system design and monitoring software.	Supply Chain: Established partnerships with suppliers of renewable technology components (e.g., solar panels, batteries).
DevelopmentStrategy to develop this capability (if needed)	Pilot Project: Implement a small-scale prototype in a controlled environment to test feasibility and refine technology.	Training: Upskill teams on solar and wind power management, energy storage, and IoT monitoring.	Incremental Development: Roll out larger-scale solutions based on pilot results and real-world feedback
Requirements and cost to develop(high/med/low)	Solar Panels – Medium cost (dependent on the number of panels and efficiency).	Wind Turbines– Medium cost (based on turbine specifications and size).	IoT Monitoring System – Low to Medium cost (depends on the level of sophistication required).

External Sources of Expertise(potential partnership)	Renewable Energy Consultants: For design and efficiency improvements.	Battery Manufacturers: Partnerships with firms specializing in high-capacity and durable energy storage.	IoT Solution Providers: Companies with expertise in integrating IoT monitoring systems. Universities or Research Institutes: Collaborations on renewable energy research and pilot programs.

5.2 EVOLVING THE PROCESS FOR DELIVERY

KeySolution Component	Workflow/ProcessNeededtoImplementtheSolution					
(InputYourSolution Component 1)	Process1 (Header)	Process2 (Header)	Process3 (Header)	Process4 (Header)	Process5 (Header)	Process6 (Header)
	Process1(De- scription)	Process 2 (Description)	Process 3 (Description)	Process 4 (Description)	Process 5 (Description)	Process 6 (Description)
(InputYourSolution Component 2)	Design Requirements Gathering	Collect input from users and stakeholders to define the suction power requirements and features for an improved suction system.	Ergonomic Research	Gather feedback from potential users on performance and comfort, adjusting designs as needed.	User Feedback Collection	Energy Capacity Analysis
(InputYourSolution Component 3)	Prototype Development	Develop and build initial prototypes with enhanced suction capabilities for testing and iteration.	Handle Prototype Development	Make final adjustments to suction power and other product aspects based on feedback and testing results.	User Feedback Collection	solar Selection

(InputYourSolution Component 4)	Testing and Quality Assurance	Conduct rigorous testing to ensure the suction mechanism meets quality standards and performs effectively across surfaces.	Usability Testing	Make final adjustments to suction power and other product aspects based on feedback and testing results.	Market Release Preparation	Turbine Design

5.3 IMPACT EVALUATION INDICATORS

Criteria	Indicators&Measurement	Stakeholders
Social Value Creation	• Community Well-being - Measure improvements in quality of life through surveys or public health data. • Economic Empowerment - Track job creation, income levels, and access to resources.	• Beneficiaries (e.g., local communities, target groups) • Funding Partners (e.g., government bodies, NGOs)

Stakeholder Satisfaction	• Beneficiary Satisfaction - Conduct satisfaction surveys or feedback sessions. • Stakeholder Engagement - Measure involvement rates and engagement levels through participation metrics.	• End Users/Beneficiaries • Partners/Investors
Solution Sustainability	• Long-term Financial Viability - Assess funding, revenue streams, and cost-efficiency. • Environmental Impact - Measure carbon footprint, resource use, and waste management.	1. Local Government and Policy Makers 2. Community Leaders

Solution Scalability	• Adaptability to New Markets - Track expansion potential and replication success rates. • Cost Efficiency of Scaling - Measure costs per beneficiary in new locations.	• Project Managers/Implementers • Investors/Funders
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5.4 ACTION PLANNING TO ADVANCE THE DESIGN CHALLENGE PROJECT

Idea What idea for implementation	Objectives Why is this idea important? Values and benefits	Responsibility Who will lead this?	Implementation How will this be Implemented?	Resources What capability and Resources are needed?	Completion When will this be completed?
Renewable charging station using solar and wind turbines.	Outline why this project is valuable, e.g., promoting sustainable energy use, reducing carbon footprint, and providing a reliable power source for electric vehicles.	Assign a project leader and key team members responsible for each stage.	Describe how the charging station will be set up, including steps like site selection, technology sourcing, and infrastructure setup.	Identify what you need in terms of technology (solar panels, wind turbines), financial investment, and technical expertise.	Set target dates for each phase, such as prototyping, testing, and final deployment.
Add a rechargeable battery option for cordless convenience	Enhance usability by making the device cordless for flexibility in use and accessibility	Engineering Team Lead	Source a rechargeable battery, adjust device design to accommodate battery	Battery supply, electrical engineering support, access to testing labs for durability testing	Initial testing completed within 3 months
Integrate RFID for enhanced access	Improve market appeal by ensuring the device captures energy effectively	Health and Safety Consultant, Engineering Lead	Research solar compatibility, integrate design changes, and conduct wind capture tests	testing equipment, collaboration with safety experts	Testing and approval within 4 months

5.5 IDENTIFYING QUICK WIN

What is this Quick Win (1) about?	Installing the first small-scale prototype.
What are the success indicator(s)? How would it (these) be measured?	Define measurable goals, such as power output or user satisfaction rates.
What are there sources/staff trainings needed?	List any additional equipment or training needed.
Who will lead this Quick Win implementation?	Project Manager: A dedicated project manager can oversee the implementation process, ensuring timelines, resources, and stakeholder communication are managed effectively. Product Development Lead: This person will be responsible for the technical aspects of the charging station, guiding any necessary design or functionality changes. Marketing Specialist: They can help assess market needs, communicate the Quick Win features to customers, and support promotional activities.
What are the key steps needed to implement this Quick Win? What is the timeline till completion?	Identify Improvement Area: - Analyze current performance (e.g., generated power, ease of use). - Gather feedback from users or customers. Research Solutions: - Investigate possible enhancements (e.g., better turbine, ergonomic handle).

	• Compare costs and feasibility. • Design the Improvement: • Create a simple design or plan for the enhancement. • Ensure it's easy to implement without major redesigns. • Week 1: Identify improvement area and gather feedback. • Week 2: Research and design potential solutions. • Week 3: Develop a prototype. • Week 4: Conduct user testing and gather feedback. • Week 5: Finalize the design and prepare for production. • Week 6: Launch the improved product.
When will be the status or progress update?	For most charging stations, manufacturers typically provide status or progress updates on production or improvements through their official website, customer support, or via email newsletters. If you are referring to a specific model or brand of vacuum duster, please let me know, and I can help you find specific information or guide you on where to check for updates.
When will this be completed?	• Identify Common User Pain Points: Quick improvements can often come from fixing the most common complaints, such as noise level, generating power, or battery life. • Enhance the Design for Portability and Convenience: Adding or optimizing carrying handles, button placements, or accessories could lead to a quick win.
How would the Success be communicated?	• Ease of Use: Demonstrating that the product is user-friendly right out of the box. • Versatility: Highlighting its effectiveness on different places (e.g., colleges, public places) to show it's more versatile than a standard charging station.

5.6 CONCEPT

Design Challenge	• Provide reliable, renewable charging options for people on the go, especially in public or remote areas. • Reduce reliance on traditional power sources for charging mobile devices and laptops.	Solution Concept	• Develop a portable or semi-permanent charging station powered by solar panels and wind turbines, specifically designed for mobile devices and laptops. These stations can be deployed in public parks, outdoor workspaces, university campuses, or remote work locations.
• Persona: • Name: THANUJA • Age: 25 • Occupation: Busy professional and parent • Lifestyle: Values efficiency and cleanliness; minimal spare time • . •			• Value Propositions to Organizations/ Enhances public spaces and facilities by providing useful amenities.
• Deep Needs: • Access to charging options in areas where traditional outlets are not available. • Reducing environmental impact through renewable energy for small-scale device charging. • Convenience and ease of use without complex setup or maintenance. •	• Value Propositions to Target Users: • Portable, renewable, and easy-to-use charging options for phones and laptops. • Ability to stay connected and powered on the go, without relying on fossil-fuel-powered outlets. • Clean, sustainable energy solution that aligns with eco-conscious lifestyles.. •	• Demonstrates a commitment to sustainability and innovation in public infrastructure. • ☑ Reduces energy consumption	

5.6 CONCEPT

• Gains: • Reliable and convenient access to charging for mobile devices and laptops in public spaces. • Contribution to a greener planet by using renewable energy. • Cost savings on electricity for users in the long term. • ?	• UserNeed(Problem) Solvers: • Solar panels and small wind turbines to generate clean energy. • Built-in battery storage to hold surplus energy for low-sun or low-wind conditions. • Compatibility with multiple devices, including USB and Type-C ports for mobile devices and laptops		from traditional grids, contributing to greener city initiatives.needed ed compared to traditional dusting methods.
• Pains: • Limited availability of charging infrastructure in outdoor and remote areas. • Dependency on conventional power sources, leading to a higher carbon footprint. • Frustration from low battery anxiety, especially in public spaces or while traveling.	• GainCreators: • Energy independence with solar and wind power. • Convenient and eco-friendly charging solution for small electronics. • User-friendly interface for easy charging access. • ?	• PainRelievers: • Eliminates the need to search for wall outlets in public spaces. • Offers continuous power options even in off-grid locations. • Reduces dependence on fossil fuels for small electronics charging.	• Improved Air Quality: By capturing fine dust particles that would otherwise become airborne, it promotes a cleaner environment, potentially reducing allergens and enhancing overall indoor air quality. • Versatility: Many vacuum dusters come with interchangeable attachments suitable for hard-to-reach areas, upholstery, electronics, and more, offering a single solution for diverse cleaning needs. • Cost Savings on Supplies: Unlike disposable dusters that require frequent replacement, vacuum

5.6 CONCEPT

			dusters can be used repeatedly with minimal maintenance, resulting in long-term savings.
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5.7 M–A–R–S FRAMEWORK

Use the MARS framework to understand the people's behavior in the face of the change and innovation Motivation It is about the Why/the Will to change					
		Ability	Role	Systems	
		It is about the Tools and Skills	It is about defining Roles/ToRs	It is about the Support system (Resources)	
Engage team in conversation to connect and to empathize. Listen to welcome the truth and to gain trust	Think	Why: Help the team understand the importance of sustainable solutions for everyday needs, emphasizing the impact of small actions, like charging devices sustainably.	Accountability: Assign clear roles like project lead, technical expert, design lead, and operations manager.	Support System (Resources): Ensure that the necessary resources (financial, human, technological) are in place to support the change. This could involve securing funding for new equipment, hiring additional staff with specialized skills, or implementing software to track progress.	
	Feel	Will to Change: Inspire excitement and a sense of purpose around creating a product that contributes to sustainability and serves a tangible public need professional	Feedback Mechanisms: Foster a sense of accountability and ownership for each role. Testing sessions.	Conversations: Facilitate team meetings to keep everyone aligned and encourage collaboration across roles.	

5.7 M–A–R–S FRAMEWORK

		growth or recognition), organizational goals (like market competitiveness), or user feedback that highlights the need for improvement.			
	Do	Tools and Skills: Plan for resource allocation, such as budget management for prototyping and deployment.	Set up regular check-ins to track progress, address challenges, and make sure all necessary support is available.	Conversations: Ensure access to materials such as solar panels, wind turbines, storage batteries, and other required components.	

Vision(Reasons)for change	Staff Engagement	Communicate vision for change	Implementation Plan	Empower people for change	Create Quick Wins
• Environmental Impact: Transitioning to renewable energy reduces carbon emissions and promotes sustainability. • Cost Efficiency: Utilizing solar panels and wind turbines can decrease energy costs over time. • Innovation Leadership: Position the organization as a leader in adopting green technologies. • Community Engagement: Enhances the organization's image and relationship with	• Involvement in Planning: Include staff in the decision-making process regarding the design and placement of charging stations. • Feedback Mechanisms: Establish regular meetings and surveys to gather input and address concerns. • Training Sessions: Offer training on how to use and maintain the new charging stations effectively.	• Kick off Meetings: Organize an event to announce the project, outlining its benefits and the timeline for implementation. • Regular Updates: Send newsletters or emails to keep staff informed of progress, upcoming milestones, and how they can contribute. • Visual Aids: Create posters and infographics highlighting the vision and benefits of the renewable charging stations.	• Project Timeline: Establish a clear timeline with key milestones, such as site selection, installation, and launch. • Resource Allocation: Determine budget, equipment, and personnel needed for the project. • Partnerships: Collaborate with renewable energy vendors and local government for support and resources.	• Assign Champions: Designate team members as change champions to advocate for the project and assist peers. • Skill Development: Provide opportunities for staff to learn about renewable energy technologies and project management. • Encourage Innovation: Create a platform for employees to propose additional ideas related to sustainability.	• Pilot Projects: Start with a few charging stations to demonstrate effectiveness and gather user feedback. • Showcase Successes: Highlight the energy savings and user satisfaction from the pilot projects to motivate continued support. • Celebrate Achievements: Recognize teams or individuals who contributed significantly to the project to reinforce commitment and enthusiasm.

the community by promoting eco-friendly initiatives.					
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CONCLUSION

In conclusion, applying design thinking to develop renewable charging stations for mobile and laptop charging has enabled us to create a solution centred on user needs, environmental impact, and technical feasibility. By following the design thinking process—empathizing with users, defining the problem, ideating solutions, prototyping, and testing—we have arrived at a sustainable and innovative approach using solar panels and wind turbines.

This solution not only offers a convenient, off-grid charging option but also promotes environmental awareness and aligns with sustainability goals. Initial prototypes and feedback cycles have shown that this renewable charging station design meets users' needs for convenience, accessibility, and eco-friendly energy. Moving forward, continued iterations and user feedback will allow us to refine and scale the solution, ultimately making it a valuable and sustainable addition to our environment-conscious infrastructure.